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The link between DFA portfolio performance, AI financial management, GDP, government bonds growth and DFA trade volumes

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Abstract
The intention of the article is to delve into digital financial assets (DFA) portfolio price growth, financial management based on artificial intelligence (AI), GDP, government bonds growth (GBG) and DFA trade volumes within the highly dynamic market environment of Russia. This study aims to make a significant contribution by providing methods (statistical methods and fuzzy logic) for investors to identify and select the most optimal long-term portfolio from the pool of 218 digital financial assets which is available in the Russian market. It is important to acknowledge the fact that companies listed as digital financial asset operators often offer multiple classes of these assets for trading, and as such, investors are only able to trade the floating digital financial assets. Nevertheless, owing to the relatively short history of digital financial assets and the presence of significant standard deviations in DFA returns within the market, there still exists a level of uncertainty regarding the sustainability of the price premium. In actuality, considering the impact of volatility, it becomes evident that the price premium did not exhibit statistical significance throughout the sample period. So, the novelty of study is creation new effective tools for researching the effectiveness of portfolio management for time series includes 416 daily observations for the period March, 2022 - October, 2023.

Keywords Digital financial assets · Fintech · Decision making · Blockchain innovation

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1 Introduction

DFAs are digital rights, including monetary claims, the possibility of exercising rights on equity securities, the right to participate in the capital of a non—public joint-stock company, the right to demand the transfer of equity securities, which are provided for by the decision on the issue of DFAs. According to the law, digital financial assets are not a means of payment. The issue (entry into the information system of information about the crediting of the DFA by their first owner) and the circulation of digital financial assets in information systems are regulated by Federal Law On Digital Financial Assets, Digital Currency and on Amendments to Certain Legislative Acts of the Russian Federation (Mikhaylov 2021; Majeed and Jamshed 2023; Rapposelli et al. 2023; Singh et al. 2023).

Along with the DFA, digital rights may also be issued in information systems, including both the DFA and the right to demand the transfer of a thing, exclusive rights to the results of intellectual activity and rights to use the results of intellectual activity, performance of works and provision of services. Such digital rights can be called hybrid. The information system operator is responsible for ensuring the smooth functioning of information systems in which the DFA is issued and recorded. It also allows users to access the information system, interacts with the authorities on the provision of information and the execution of court decisions. In addition, the operator of the information system has the opportunity to ensure transactions with the DFA and settlement of transactions with the DFA using a nominal account (the right to settle transactions is reflected in the corresponding section of the register of operators) (Amoako et al. 2021; Zhang and Lu 2021; Hemanand et al. 2022).

The paper uses tools of Post-Modern Portfolio Theory for enhancing efficiency and mitigating challenges related to mismanagement in DFA market.

Paper objective is the link between the digital financial assets (DFA) portfolio price growth, financial management based on artificial intelligence (AI), GDP, government bonds growth and DFA trade volumes of DFA within the highly dynamic market environment of Russia at the beginning of complex and fascinating process of digital financial asset evolution in Russia's highly dynamic market environment. In the traditional financial sector, the yield on short-term DFAs corresponds to the established investment terms. However, it should be noted that due to the rare issuance of short-term DFAs by the Russian Ministry of Finance, a strategic alternative has emerged. This alternative involves the use of long-term or medium-term DFAs with a remaining maturity of about one year, which is an effective substitute. (Paul et al. 2020; Flores et al. 2020; Sircar et al. 2021).

It is clear that DFAs significantly outperformed short-term DFAs in terms of financial performance and earnings between Febryary, 2022 and 2023. Compared to the return on equity over the same period, the yield on long-and medium-term DFAs is noticeably and significantly lower than the return on equity, which is characterized by a relatively higher yield. It is noteworthy that from 2022 to 2023, long-term DFAs performed exceptionally well, surpassing the yields of long-term and short-term DFAs. It is noteworthy that the cumulative annual growth rate of long-term DFAs was 4.94%, which indicates their potential profitable investment potential. Conversely, over the same period, an initial investment in DFAs (Abraham 2022; Adekoya et al. 2022).

The reason for this discrepancy is that almost all long-term DFAs analyzed in this context were issued by state and local state-owned enterprises or agencies, so the risk of default is almost negligible. In addition, it is clear that DFA indices show significantly greater volatility compared to fixed income indices and inflation. Conversely, the latter group tends to follow a more linear upward trajectory. Given the high volatility of wealth indices, it is particularly interesting to observe their impact on the geometric averages of DFAs, especially when compared to their arithmetic counterparts. This comparison shows that the geometric averages of DFAs are noticeably lower than their arithmetic counterparts (Aslan et al. 2023; Avdjiev et al. 2022).

In conclusion, main research question should be recognized that both long-term DFAs are characterized by a significant level of volatility, although below the volatility of DFAs. In addition, the lower level of volatility in medium-and short-term DFAs once again underlines their stability. In addition, it is necessary to consider the overall volatility of inflation, as it has had a noticeable impact on the financial landscape. Finally, comparing the geometric and arithmetic averages of DFAs highlights the detrimental effect of high volatility on long-term equity investors. The average return on DFAs of these companies adjusted for inflation was very remarkable. We will find that average DFA returns have declined significantly: the average return on DFAs adjusted for inflation was only about 1.64% per year (Draganidis 2023; Eichengreen et al. 2023).

In addition, it is worth noting that over the same period, medium-and long-term DFAs showed more favourable performance indicators. Finally, it is extremely important to shed light on the dynamics of long-term DFAs. Obviously, these DFAs were not as good as other investment options, as their inflation-adjusted returns were only 2.66% per year between 2022 and 2023. While this figure isn't as impressive as the returns on DFAs it still provides an opportunity for investors looking for a more conservative and reliable way to invest. Therefore, it is recommended that you carefully analyze and consider the various performance indicators and risk factors associated with each investment option before making any financial decisions. The marked discrepancy between the arithmetic mean return and the geometric return of DFAs can be attributed to their inherent high volatility. As for DFAs, their average annual yield, adjusted for inflation, varies over a wide range of values. This indicates a certain level of stability in these DFAs. However, the impact of inflation on DFA yields is relatively limited, primarily due to their inherent volatility. It is important to note that after adjusting for inflation, the volatility of short-term DFA yields increases (Azam et al. 2022; Dinçer et al. 2022).

The subsequent segment comprises an assessment of the existing literature, delving into the various studies and scholarly works that have been conducted in the field. This thorough examination aims to provide a comprehensive overview of the current state of knowledge and identify any gaps or areas in need of further investigation. The methodology, which delineates the systematic approach employed in this research endeavour, is expounded upon in the subsequent section. This detailed elucidation of the research methodology offers transparency and allows for the replication of the study by other researchers, thereby contributing to the advancement of knowledge in the field. The analysis findings, which are the outcomes of the rigorous application of the chosen methodology to the collected data, are comprehensively depicted in the ensuing section. These results are presented with an emphasis on their significance and implications, shedding light on their relevance to the research questions and objectives. The discourse and conclusion, encapsulating the critical reflection and final summation of the research process and outcomes, are articulated in the subsequent sections, offering a thoughtful synthesis of the key findings and their broader implications for the field. This reflective and conclusive section provides a comprehensive wrap-up of the study, offering insights into its broader significance and potential avenues for future research.

2 Literature review

2.1 DFA research

The volatility of inflation-adjusted returns across the six asset classes mentioned above shows identical patterns and values compared to nominal returns. An analysis of returns over a five-year rolling period shows an intriguing pattern: in more than 80% of the 20 sample periods, all assets showed positive returns (Ghazani and Jafari 2021; Guo et al. 2022; Geyikçi and Özyıldırım 2023).

Statistical analysis of historical returns on assets provides valuable information about their characteristics, such as average returns, risk assessment based on yield volatility, and simultaneous movement of asset returns. It is worth noting that when comparing the history of DFAs of companies with the history of long-term DFAs, the history of the former is usually longer. Overall, the performance of DFAs has shown a higher level of volatility compared to long-term DFAs throughout their respective history. It is noteworthy that the volatility of DFAs was especially pronounced in the early years, when only a limited number of DFAs were sold and there were no restrictions on daily price fluctuations. This increased volatility eventually peaked towards the end of 2022. Subsequently, in 2023, volatility entered a relatively moderate phase. However, from 2022 to 2023, volatility increased again, although it remained below the extreme levels seen in previous years. DFA market volatility is usually more moderate and characterized by a certain stability in the long term. From 2022 to early 2023, the situation with long-term DFAs was characterized by an increased level of volatility. This influx, in turn, contributed to a period of increased volatility in the DFA market. When calculating the monthly profitability of the entire market, a comprehensive approach is used. First of all, the return on individual DFAs for each month is carefully calculated. After that, the portfolio's return is then estimated based on the weight of each individual DFA in the portfolio. This thorough process provides a comprehensive and accurate assessment of the monthly market return, which allows you to make informed decisions and carry out strategic planning, which is extremely important in any investment project (Hirota et al. 2022; Kafka et al. 2022).

Some asset returns can be classified as random, while others can be considered unpredictable or subject to certain trends and periodicity. The predictability or lack of such return on assets can be reflected in their sequential correlations or autocorrelations. From a theoretical point of view, a sequence with a high autocorrelation coefficient of 1 is considered completely predictable. On the other hand, a sequence with coefficients close to zero is considered almost completely random, which makes it difficult to predict. In general, DFA market volatility tends to decrease, with notable exceptions during certain periods of increased volatility. Calculating monthly market returns requires careful evaluation of individual DFA returns and then determining portfolio returns. Finally, the predictability of asset returns depends on various factors, including their correlations and autocorrelations, which allows us to understand in detail their potential predictability (Mensi et al. 2023; Metaxas et al. 2023; Mikhaylov 2023).

Long-term DFAs, long-term DFAs, and medium-term DFAs are considered relatively less risky investment options, which consequently leads to lower average yields. Interestingly, the annual yield on long-term DFAs has a significant negative constant correlation (-0.47), which indicates an inverse relationship between the yield and other significant factors. In contrast, the yields on long-term and medium-term DFAs have only a slightly positive constant correlation (0.02 and 0.02, respectively), which indicates a moderate positive relationship between the yield and other significant variables. Notably, this standard deviation serves as a useful indicator of the extent to which these interest rates change over time and provides investors with valuable information. Moreover, the strong correlation seen in short-term DFA yields highlights the steady nature of short-term interest rates and, consequently, their constancy over time. This correlation, as shown by empirical data, once again highlights the relationship between short-term interest rates and their impact on the yield of these DFAs (Moiseev et al., 2023).

2.2 Contribution in post-modern portfolio theory

In addition, it is worth recognizing that the correlation between the returns of various assets reflects the overall risks they face. By studying these correlations, investors gain a full understanding of the overall risks associated with different assets. Such analytics are particularly valuable because they shed light on the overall risks inherent in investment portfolios, and thus enable investors to effectively manage and address potential vulnerabilities. It is important to note that these general risks are of considerable interest and concern to investors, as they indicate systemic factors to which assets are exposed. By identifying and assessing these systematic risks, investors can make informed decisions and take appropriate measures to protect their investment portfolios (Li et al. 2022; Martínez et al. 2023; Guo and Zhao 2023; Xiong et al. 2023; Alexeev et at., 2023).

In investment theory, it is widely believed that systematic risks are ultimately related to the expected return on these assets. This belief is based on the understanding that assets exposed to higher systemic risks should generate higher returns, compensating investors for additional risks. By recognizing the relationship between systematic risks and expected returns, investors can make informed decisions about the allocation of their investment capital. Moreover, this understanding underscores the importance of carefully analyzing and understanding the systematic risks associated with various assets, as they play a key role in shaping investment strategies and results (Naeem et al. 2023; Ospina-Forero and Granados 2023; Reus and Sepúlveda-Hurtado 2023).

In short, short-term DFAs represent a virtually risk-free investment opportunity with moderate returns. The standard deviation of these companies ' returns reflects the impact of short-term interest rate fluctuations, rather than the inherent risks of returns. In addition, the strong correlation between the yields of these DFAs reflects the steady nature of short-term interest rates. The correlation between the returns of different assets highlights the common risks they face and provides investors with valuable insights. Of particular interest are general risks that include systemic elements that are believed to be related to expected returns. Understanding and managing these systematic risks is crucial for investors to effectively manage the investment landscape (Li et al. 2022; Martínez et al. 2023).

The paper has 2 hypothesizes:

Hypothesis 1 The impact of LGBG on the LDFAPG is negative and significant.

Hypothesis 2 The impact of LGBG on the LDFAPG is positive and significant.

This means that as DFA premiums increase, so do default and maturity premiums. In addition, DFA premiums also have a positive impact on inflation. This means that as DFA premiums rise, inflation also tends to rise. In addition, investors express concern about changes in risks, which can be estimated by examining the standard deviation or volatility of returns over different time periods. This risk metric allows investors to assess potential fluctuations in returns and adjust their investment strategies accordingly (Li et al. 2022; Martínez et al. 2023).

3 Data and methodology

In this part of the study, necessary information is given with respect to the data used in the analysis and proposed methodology.

3.1 Data

The data used in the analysis process is obtained from Bank of Russia. It includes 218 digital financial assets issued by Russian operators on 07.11.2023 (Tables 1 and 2; Fig. 1).

Time series includes 416 daily observations for the period March, 2022 - October, 2023 which is enough for efficient modelling and creation new effective artificial intelligence tool (Paul et al. 2020; Flores et al. 2020; Sircar et al. 2021; Amoako et al. 2021; Zhang and Lu 2021; Hemanand et al. 2022).

First, it is worth noting that an increase in long-term DFA returns cannot be attributed solely to higher systemic risk. While this relationship may not be fully explicable, empirical research consistently shows that the return on DFAs exceeds the expected return on DFAs (Tables 3, 4 and 5). Second, the difference in annual DFA returns is strongly correlated, suggesting that historical annual returns can be very useful for predicting future returns. This robust correlation, commonly referred to as autocorrelation, has been found to have no empirical significance in the context of DFAs and most other capital markets (Yayla et al. 2023; Yüksel and Dinçer 2023; Zhang et al. 2023).

It is important to note that while the annual returns of DFAs show a slight constant correlation, portfolios show a slight negative autocorrelation. DFAs tend to be exposed to a higher level of risk, it is important to recognize that their potential to generate substantial returns can serve as a means of partially offsetting this inherent risk. The Capital Asset Pricing Model (CAPM) is a widely recognized and accepted system that serves as a starting point for determining the risk premium associated with a given asset, which is essentially a form of compensation for the level of risk it represents. To quantify the systematic risk of an asset, it is common to measure its beta, which is essentially a numerical representation of the asset's volatility relative to the market as a whole. It is worth noting that a beta value greater than 1 indicates that this security is riskier than the market as a whole. According to CAPM principles, investors are rightfully rewarded for taking on additional systematic risk. However, when analyzing the historical dynamics of portfolios, in particular in the

Table 1 Most high-volume traded digital financial assets in Russia

Number	DFA name	Issuer	Coupon Code	Announced volume
1	RUSSIAN RAILWAYS, FRN 13dec2023, RUB (CFA)	RUSSIAN RAILWAYS	1 payment - Key rate of the Central Bank of the Russian Federation + 0.95% per annum	15 000 000 000
2	Alfa-Bank, 6dec2023, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	3 200 000 000
3	Alfa-Bank, 4mar2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	3 000 000 000
4	Alfa-Bank, 13dec2023, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	2 000 000 000
5	Forte Home GMBH, 30apr2025, RUB (CFA)	Forte Home GMBH	The amount of income payment is determined in accordance with the formulas specified in clause 2.11 of the Decision to issue a CFA	1 500 000 000
6	Automated cash settlements	Automated cash settlements	11.6547% per annum (calculated)	1 000 000 000
7	Alfa-Bank, 10% 13may2024, RUB (CFA)	Alfa-Bank	10%	1 000 000 000
8	Alfa-Bank, 12feb2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
9	Alfa-Bank, 15jan2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
10	Alfa-Bank, 15jan2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
11	Alfa-Bank, 19feb2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
12	Alfa-Bank, 26feb2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
13	Alfa-Bank, 26feb2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
14	Alfa-Bank, 29feb2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
15	Alfa-Bank, 7oct2024, RUB (CFA)	Alfa-Bank	Determined in accordance with clause 2.11. of the Decision to issue a CFA	1 000 000 000
16	MTS, FRN 22aug2023, RUB (CFA, NDM_1)	MTS	1–2 payments - Key rate of the Central Bank of the Russian Federation + 1.72% per annum	1 000 000 000
17	Sberbank Factoring, 6.9054% 30sep2022, RUB (CFA, 9,162,996 C)	Sberbank Factoring	1 payment – 6.9054% per annum (calculated)	1 000 000 000

Source Bank of Russia

Russian markets, it becomes obvious that the decline in profitability observed in the lower deciles cannot be fully explained by the CAPM model alone. This discrepancy is further compounded by the residual excess of the CAPM index, which becomes more significant as companies move from the first decile to the companies in the tenth decile (Wang and Zhou 2022; Wang and Zhong 2022; Srbova et al., 2023).

Table 2 Digital financial assets operators in Russia

Date of inclusion	Operator	Volume of issue, million rubles	Number of Crypto issues in circulation, pcs
02/03/2022	Atomize	1150	2
17/03/2022	Sberbank	307	13
17/03/2022	Lighthouse	751	4
02/02/2023	ALFA-BANK	750	1
03/09/2023	Distributed Registry Systems	0	0
15/06/2023	Tokens – Digital Investments	0	0
15/06/2023	EUROFINANCE MOSNARBANK	0	0
22/06/2023	SPB Exchange	0	0
27/07/2023	Blockchain Hub	0	0
03/08/2023	National Settlement Depository	0	0

Source Bank of Russia

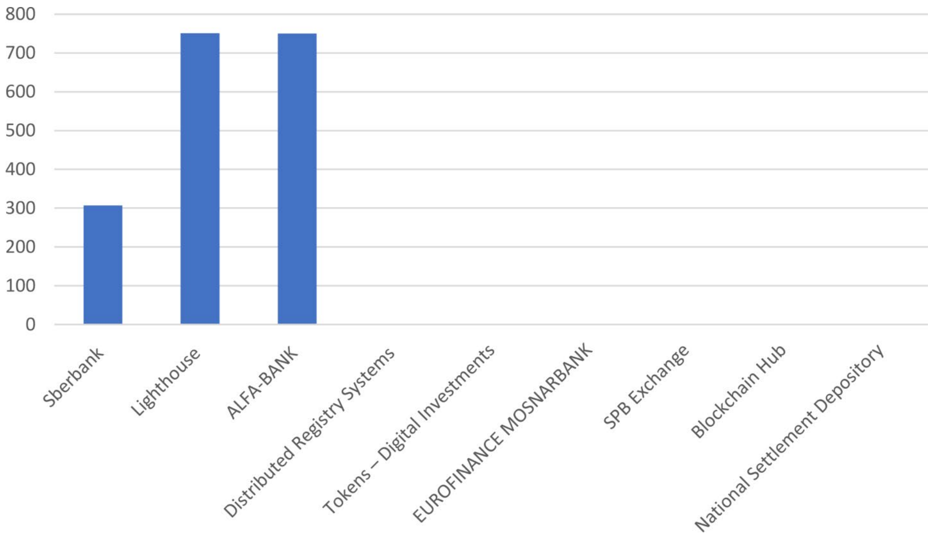

Fig. 1 Digital financial assets volumes in Russia, mln. Rubles. *Source* Bank of Russia

3.2 Methods

The present study examines the connections between DFA portfolio price growth, financial management based on artificial intelligence (AI), GDP, Government bonds growth and DFA trade volumes. The analysis utilizes annual time series data from DFA in Russia for the period from 2022 to 2023. All the data for the variables is obtained from the Bank of Russia. Drawing upon existing literature, the general econometric model is structured as follows:

$$\mathrm{DFAPGt} = \gamma 0 + \gamma 1 LFMt + \gamma 2 LGDPt + \gamma GBGt + \gamma 4 LTRt + \varepsilon t \tag{1}$$

Table 3 Benchmarks of long-term digital financial assets in Russia

Business Investment, 6.9091% 1dec2033, RUB (CFA, 2DA857D7)	Business Investments	80 000 000	01.12.2033
Business Investment, 6.9204% 1dec2033, RUB (CFA, C228376D)	Business Investments	80 000 000	01.12.2033
Business Investment, 6.941% 1dec2033, RUB (CFA)	Business Investments	80 000 000	01.12.2033
Business Investment, 6.9448% 31oct2033, RUB (CFA, 444723A4)	Business Investments	142 000 000	31.10.2033
Digital Assets, 8nov2028, RUB (CFA, MINETOKEN_44)	Digital Assets	150 000 000	08.11.2028
Digital Assets, 8nov2028, RUB (CFA, MINETOKEN_45)	Digital Assets	150 000 000	08.11.2028
Digital Assets, 8nov2028, RUB (CFA, MINETOKEN_46)	Digital Assets	150 000 000	08.11.2028
Digital Assets, 8nov2028, RUB (CFA, MINETOKEN_47)	Digital Assets	150 000 000	08.11.2028
Digital Assets, 8nov2028, RUB (CFA, MINETOKEN_48)	Digital Assets	150 000 000	08.11.2028
Digital Assets, 3nov2028, RUB (CFA, MINETOKEN_43)	Digital Assets	150 000 000	03.11.2028
Digital Assets, 1nov2028, RUB (CFA, MINETOKEN_42)	Digital Assets	150 000 000	01.11.2028

Source Bank of Russia

Table 4 Benchmarks of middle-term digital financial assets in Russia

Expobank, 26nov2026, RUB (CFA)	Expobank	1 975 000	26.11.2026
PR-Leasing, 14% 2oct2026, RUB (CFA, NDM_13)	PR-Leasing	50 000 000	02.10.2026
Airplane Plus CFA, 31dec2025, RUB (CFA, NDM_2)	Airplane Plus CFA	2 430 423	31.12.2025
Airplane Plus CFA, 31dec2025, RUB (CFA, NDM_10)	Airplane Plus CFA	514 128	31.12.2025
Basis Plus, 0.485372% 31jul2025, RUB (CFA, 2676D2CF)	Basis Plus	500 000	31.07.2025
Forte Home GMBH, 30apr2025, RUB (CFA)	Forte Home GMBH	1 500 000 000	30.04.2025

Source Bank of Russia

Table 5 Benchmarks of short-term digital financial assets in Russia

Radio City, 0.0289% 31jan2024, RUB (CFA, A2C864BB)	Radio City	3 000 000	31.01.2024
Radio City, 0.0308% 31jan2024, RUB (CFA, 8,339,511 A)	Radio City	2 500 000	31.01.2024
Radio City, 0.0262% 31jan2024, RUB (CFA, 506106F3)	Radio City	1 400 000	31.01.2024
Alfa-Bank, 29jan2024, RUB (CFA)	Alfa-Bank	700 000 000	29.01.2024
Pioneer Group, 15% 19jan2024, RUB (CFA)	Pioneer Group of Companies	100 000 000	19.01.2024
Alfa-Bank, 13% 16jan2024, RUB (CFA)	Alfa-Bank	100 000 000	16.01.2024
Alfa-Bank, 16jan2024, RUB (CFA)	Alfa-Bank	100 000 000	16.01.2024
Alfa-Bank, 15jan2024, RUB (CFA)	Alfa-Bank	1 000 000 000	15.01.2024
Alfa-Bank, 15jan2024, RUB (CFA)	Alfa-Bank	1 000 000 000	15.01.2024
NTS Gradient, FRN 10jan2024, RUB (CFA, NDM_12)	NTS Gradient	100 000 000	10.01.2024

Source Bank of Russia

In this equation, DFAPG represents DFA price growth, FM represents financial management represented by AI, measured through patent applications. AI is a strategic tool that enhances the benefits of management. Its application in the natural or mineral resource market provides significant advantages by improving financial management techniques, accuracy measures, and long-term returns. GDPPC represents Gross Domestic Product per capita (GDP), GBG represents government bond price growth, and TR represents trade, which are important control variables in the literature that have a dominant impact on DFA portfolio management. L represents the log transformation, subscription t represents the time period, while γ0 represents the intercept measuring the impact on MRR in the absence of other independent variables. γ1 to γ4 represent the parameters indicating the impact of an associated variable on the MRR. εt indicates the error term.

With regards to the empirical methodology, the decision is made considering the nature of the data. Since the data is a time series, the first and most crucial step is to ensure the stationarity of the data. The Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests are used for this purpose. Both the ADF and PP tests are widely used to assess the stationary feature of series in time series data. The ADF test is an extended version of the Dickey-Fuller test designed to handle higher order processes. By using the autoregressive moving average (ARIMA) model, this test addresses the issues of serial correlation and heteroscedasticity. The PP test, proposed by Phillips and Perron (1988), also provides reliable results when dealing with serial correlation and heteroscedasticity in the error term. The KPSS test assumes the null hypothesis that the series is stationary and is practically the opposite of the ADF and PP test interpretations. The test is applied to the models, and the general form is as follows:

$$\Delta DFAPGt = \alpha + \beta t + \gamma LDFAPGt - 1 + \delta 1 \Delta LDFAPGt - 1 + ... + \delta p - 1 \Delta LDFAPGt - p + 1 + \varepsilon t \quad (2)$$

In this equation, α is a constant, β is the coefficient of trend, and p is the order of lags of an autoregressive (AR) model. If α and β are equal to zero, then the model is a random walk, while β equal to zero indicates a model with a random walk and drift. The null hypothesis for the test is Ho: $\gamma=0$, and the alternative hypothesis is H1: $\gamma<0$.

The next long-run framework can be expressed in the following manner:

$$LDFAPGt = \gamma 0 + \gamma 1 LMt + + \gamma 2 LMt - + \gamma 3 LGDPPCt + \gamma 3 LEUt + \gamma 3 LTRt + \varepsilon t \quad (3)$$

Here, γ0, γ1+, γ2-, γ3+, γ4-, γ5, and γ6 represent vectors of unknown long-run parameters. LAIt+, LAIt-, LGPRt+, and LGPRt- correspond to the partial sum of positive and negative changes in LAI and LGPR, respectively.

Furthermore, the asymmetry in LFM can be described using the following equations:

$$\text{LFMt} + = \Sigma \text{j} = 1\text{t}\Delta\text{LFMj} + = \Sigma \text{j} = 1\text{t } \max(\Delta\text{LFMJ,0}) \quad (4)$$

$$\text{LFMt} - = \Sigma \text{j} = 1\text{t}\Delta\text{LFMj} - = \Sigma \text{j} = 1\text{t } \min(\Delta\text{LFMJ,0}) \quad (5)$$

These equations are based on the positive and negative partial sum decomposition, which allows for a comprehensive analysis of the asymmetry in LFM.

When asymmetric variables are present, the model captures the long and short-term implications as follows:

$$\begin{aligned}\Delta\text{LDFAPGt} = {} & \theta 0 + \tau 1 LDFAt - i + \theta 2 + LFMt - 1 + + \theta 3 - LFMt - 1 - + \\ & \theta 4 LGDPPCt - 1 + \theta 5 LEUt - 1 + \theta 6 LEUt - 1 + \Sigma\text{I} = 2\text{PJi}\Delta\text{LDFAPGt - i} + \\ & \Sigma i = 3q(\sigma i + \Delta LFMt - i + + \sigma i - \Delta LFMt - i -) + \Sigma i = 3r\alpha i \Delta\text{LGDPPCt - i} + \\ & \Sigma i = 3s\tau i \Delta LEUt - i + \Sigma i = 0t\pi i \Delta LTRt - i + \varepsilon t\end{aligned} \quad (6)$$

In this equation, the alphabets a to t represent the lag orders of the associated variables. These coefficients are constrained in order to obtain reliable findings: γ1+ = -θ2+τ1 and γ2- = -θ3-τ1. By using the ∑i=0qσi+term, an increase in LFM impact in the short run can be calculated, while the ∑i=0qσi- term allows for the calculation of a fall in LFM effect. This enables the assessment of the nonlinear impact of LFM on the LDFAPG of the DFA in Russia.

Following the equation, the error correction model (ECM) is as follows:

$$\begin{aligned}\text{LDFAPGt} = {} & \Sigma i = 2P\aleph i \Delta LDFAPGt - i + \\ & \Sigma i = 3q(\sigma i + \Delta LFMt - i + + \sigma i - \Delta LFMt - i -) + \\ & \Sigma i = 13r\alpha i \Delta LGDPPCt - i + \Sigma i = 3s\emptyset i \Delta LEUt - i + \\ & \Sigma i = 0t \cap Ci \Delta LEUt - i + \omega t ECTt - i + \varepsilon t\end{aligned} \quad (7)$$

Here, ωt represents the coefficient of the Error Correction Term (ECT).

In the NARDL framework, the long-run relationship is tested through the bound test. The null hypothesis (H0) states that there is no cointegration, while the alternative hypothesis (H1) posits that cointegration exists. This test is conducted using the Wald F-statistics.

Table 6 Details of the parameters

Parameter	Summary
DFAPG	DFA price growth,%
FM	Artificial Intelligence Financial Management
GDPPC	GDP per capita (constant 2022 US$)
GBG	Government bonds price growth,%
TR	Trade volumes of DFA (% of GDP)

Table 7 Descriptive summary

Parameter	DFAPG	FM	GDPPC	GBG	TR
Average	99.55052764	0.015443	0.08763	98.20517	0.03645
Standard error	0.006992234	0.006544	0.006754	0.011043	0.008754
Median	99.525	0.015721	0.008756	98.18475	0.025767
Moda	99.525	0.015089	0.006863	98.2995	0.026765
Standard deviation	0.142614152	0.765643	1.64534	0.225233	0.96753
Sample variance	0.020338796	0.07564	1.87670	0.05073	0.96154
Excess	0.707863238	0.875643	0.977645	0.939414	0.865436
Asymmetry	0.376251046	0.876554	0.675462	0.713794	0.853649
Interval	1.143	1.764	1.976	1.3525	1.564
Observations	416	416	416	416	416

Kq+ = ∑j = 3q ∂LDFAPGt + j/∂LFMt-1+.

Kq- = ∑j = 3q ∂LDFAPGt + j/∂LFMt-1-.

Here, as q approaches infinity, Kq + approaches γ1 + and Kq- approaches γ2-.

Finally, the asymmetric cumulative effect of a 1% change in the positive and negative partial sums of FDI changes (LFMt-1 + and LFMt-1-) is estimated as Kq + and Kq-, respectively. These estimates are obtained by summing the partial derivatives of LDFAPG with respect to the lagged positive and negative partial sums of FDI changes for a specific lag order q. As the lag order approaches infinity, Kq + approaches the long-run parameter γ1 + and Kq- approaches the long-run parameter γ2-.

4 Application

4.1 Correlation and statistical analysis

Moving on to the results, we first present a descriptive analysis of the variables. Table 6 provides a statistical summary of all the variables, including the number of observations, mean, maximum, minimum, and standard deviation. This information is essential for understanding the distribution and characteristics of the variables under investigation. For instance, the average DFAPG is 99.55, indicating a moderate level of growth in the DFA (Table 7). The minimum DFAPG value was in December 2022 suggests a period of slow growth, while the maximum DFAPG value was in June 2023 indicates a period of rapid growth in the DFA price.

The correlation among the variables of interest is presented in Table 8, which illustrates the degree of association between different variables. It is evident from the figure that there exists a moderate and positive correlation between FM and DFAPG, indicating that an increase in FM is likely to be accompanied by a corresponding increase in DFAPG. Furthermore, this correlation is statistically significant, suggesting that it is not due to random chance but rather reflects a genuine relationship between the two variables. In addition to this, the correlation coefficients for the other control variables, namely GDPPC, GBG, and TR, also demonstrate a moderate and positive association with the outcome variable DFAPG. However, it is worth noting that GBG exhibits a negative correlation with DFAPG, implying that an increase in GBG is associated with a decrease in DFAPG.

The application of artificial intelligence (AI) in the field of FM has the potential to revolutionize mineral engineering by offering various benefits such as accurate predictions, cost reduction, and future analysis. By harnessing AI, it becomes possible to leverage geological and mineralogical information to optimize the extraction process. Factors such as the duration of the extraction period, texture of the minerals, and their specific use can all be considered to enhance the efficiency of the mining operation. Moreover, AI can also have a significant impact on the financial aspects of FM, enabling better decision-making and improving market efficiency. By utilizing AI-based systems, management can make informed choices that consider the financial worth of different mineral deposits, ensuring that resources are allocated in a manner that maximizes profitability.

In the context of portfolio management, the findings of the study indicate that the first lag of a positive shock and a negative shock are significantly associated with LDFAPG. This implies that the performance of the portfolio is influenced by both positive and negative shocks, highlighting the importance of optimizing portfolio performance. Furthermore, the study reveals that activity efficiency and environmental stability also play a role in determining portfolio performance. This underscores the need for proper planning and coordination in order to achieve optimal outcomes. Therefore, the implementation of an AI-based FM system requires careful consideration and integration of various factors including the financial system, technology specialists, and extraction workers. Additionally, it is crucial to accurately assess the demand for DFA in order to ensure that resources are allocated appropriately.

In conclusion, the findings of this study suggest that AI-based FM is a valuable tool for achieving portfolio efficiency and improving risk management. By leveraging AI technology, accurate forecasting and other related analyses can be conducted, enabling better decision-making and ultimately leading to improved financial performance. However, it is important to recognize that the implementation of AI-based FM systems requires time and effort, as well as careful coordination and integration of various stakeholders. Nonetheless, the potential benefits of such systems make them a worthwhile investment for organizations operating in the field of mineral engineering.

Table 8 Correlation matrix

Parameter	DFAPG	FM	GDPPC	GBG	TR
DFAPG	1				
FM	0.654	1			
GDPPC	0.876	0.861	1		
GBG	−0.613	−0.551	−0.345	1	
TR	0.871	0.765	0.816	0.965	1

The impact of GDP per capita on the LDFAPG is both positive and significant, which suggests that a 1% increase in GDP per capita results in increase in the LDFAPG. However, when it comes to the short-run (SR), the effect of GDP per capita is quite different as it significantly reduces the LDFAPG. This can be explained by the fact that when per capita income rises in the SR, individuals tend to allocate their money towards essential goods and services. On the other hand, in the long-run (LR), people tend to prefer spending their money on luxurious items, thereby increasing the demand for gold and innovative DFA. It is important to note that these resources are also used as inputs in the production process, leading to an increase in the relative demand for DFA compared to its supply, ultimately resulting in a rise in the LDFAPG.

The results confirm hypothesis 1:

Hypothesis 1 The impact of LGBG on the LDFAPG is negative and significant.

The results do not confirm hypothesis 2:

Hypothesis 2 The impact of LGBG on the LDFAPG is positive and significant.

However, in the SR, both LGBG and its second lag exhibit a negative relationship with the LDFAPG, while the first lag shows a favorable correlation. DFA plays a crucial role in the manufacturing process and is heavily dependent on blockchain technology, which unfortunately has negative consequences for environmental sustainability. However, in the LR, there is a shift towards emphasizing the importance of environmental sustainability in DFA production, leading to a decrease in demand for DFA and subsequently lowering the LDFAPG. Additionally, as alternative and green methods of producing DFA expand in the LR, the LDFAPG also decreases due to reduced energy usage in the DFA issuing process.

The results found that the coefficient of trade exhibits a positive sign in the LR and a negative sign in the SR. Specifically, a 1% increase in trade volumes is associated with an increase in the LDFAPG in the LR, whereas in the SR, a rise in trade causes the LDFAPG to increasing.

This negative impact observed in the SR may be attributed to factors such as limited information, insufficient domestic investment or production, and a resulting lack of demand for DFA. However, in the LR, population growth and the subsequent increase in demand for DFA contribute to the rise in the LDFAPG. Furthermore, the knowledge gained and the political interest generated in the LR may further boost the LDFAPG. Lastly, the ECT (Error Correction Term) is found to be negative and highly significant, indicating that equilibrium in the LR will be restored at a speed of 75%.

5 Discussions

The level of volatility observed in equity returns consistently exceeds the level of DFA yields and inflation rates, which is a well-established and recurring trend. We should not lose sight of the fact that the level of asset volatility has undergone significant changes during the analyzed period from 2022 to 2023. However, there have been notable spikes in DFA market volatility at critical times, such as the sharp decline in Russia's DFA market in 2022, the subsequent global financial crisis, and the turbulent time of 2022. These events led to a significant increase in DFA market volatility. The dynamics of changes in the volatility of DFA yields over time have a similar trajectory: first it was increased, and then it decreased in the first half of the 2023. Subsequently, in the second half of the 2023, volatility increased sharply, but in subsequent years it stabilized at a lower level (Sapra and Shaikh 2023; Saqib et al. 2021; Soria et al. 2023).

The trajectory of inflation volatility followed a similar pattern: it was relatively high in the 2022, then it declined significantly in the first half of the 2023, there was an upswing in the second half of the 2023, and then it returned to a lower level recently. Standard deviations for a rolling period are calculated by systematically scanning a fixed-length window for each time series and determining the standard deviation of the asset class for each corresponding period. This particular approach has proven to be an invaluable tool for determining the volatility or level of risk associated with assets, considering waiting periods similar to those of real investors. Since the closing date of the sliding window serves as an indicator of volatility, it reflects realized volatility in the period preceding the aforementioned end date. Similarly, these standard deviations were relatively low between 2022 and 2023. However, it should be noted that after 2022, volatility increased again and reached its peak in 2022. After this peak, volatility began to steadily decline and continued until the first half of 2023. However, this trend has changed markedly: in the second half of 2023, after significant market shocks, the growth trajectory changed (Yayla et al. 2023; Yüksel and Dinçer 2023; Zhang et al. 2023).

The Capital Asset Pricing Model (CAPM), an analytical tool commonly used in the financial industry, is used to calculate returns that exceed the risk-free rate. It also serves as a means of comparing the aforementioned estimate with historical data. According to CAPM, the yield of a particular security consists of two separate components: the risk-free rate and the excess yield. This excess return is determined by multiplying the beta ratio (β) by the DFA's risk premium, which is essentially the compensation paid to investors for their willingness to bear market risk. The difference between the expected excess return projected by CAPM and the updated excess return is the premium amount. It is quite obvious that investing in a portfolio consisting of micro-capitalized DFAs will bring significantly higher returns compared to a portfolio consisting exclusively of DFAs. In addition, it is worth noting that the market of securities with an average capitalization is more stable compared to other categories. It is important to note that the line of the securities market, which serves as a graphical representation of the CAPM model, is determined without any adjustments to consider the size of the premium (Yayla et al. 2023; Yüksel and Dinçer 2023; Zhang et al. 2023).

The theoretical implications of the paper are the contributions to Post-Modern Portfolio Theory for enhancing efficiency and mitigating challenges related to mismanagement in DFA market. The practical implications and limitations are that the DFAs portfolio performance with AI financial management significantly outperformed short-term DFAs in terms of financial performance and earnings between Febryary, 2022 and 2023.

The future research directions are providing methods (statistical methods and fuzzy logic) for investors to identify and select the most optimal long-term portfolio from the pool of digital financial assets which is available in BRICS market.

6 Conclusions

The main contribution of this study is that methods for investor can find optimal long-term portfolio from 218 DFAs in Russian market. The paper fills the gap about the link between DFA price and Government bonds price in Russia and it found the strong connection.

Value firms, which can be defined in many different ways, are firms whose fundamental value is significantly higher than their market value. A significant amount of research has shown that these high-value companies tend to generate higher average returns than their growing counterparts both in the United States and in many international DFA markets. However, upon closer inspection, it becomes clear that this difference in performance is primarily due to exceptional estimates achieved in just a few years.

Given the fact that companies listed as DFA operators often have multiple classes of DFAs available for trading, it is important to note that investors can only trade floating DFAs. However, due to the relatively short history of the DFA and the presence of significant standard deviations in DFA returns, there remains some uncertainty about the sustainability of the price premium. In fact, given the impact of volatility, it is clear that the price premium was not statistically significant throughout the sample period.

These findings contribute to the Post-Modern Portfolio Theory in the field of portfolio management, as they establish that AI-based portfolio management is a contemporary tool for enhancing efficiency and mitigating challenges related to mismanagement in the DFA market.

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Declarations

Competing interests This research declares no competing interests.

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